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Java SE 27 & JDK 27

Class Currency

java.lang.Object
    java.util.Currency

All Implemented Interfaces:

Serializable

```
public final class Currency
extends Object
implements Serializable
```

Represents a currency. Currencies are identified by their ISO 4217 currency codes. Visit the ISO web site for more information.

The class is designed so that there's never more than one `Currency` instance for any given currency. Therefore, there's no public constructor. You obtain a `Currency` instance using the `getInstance` methods.

Users can supersede the Java runtime currency data by means of the system property `java.util.currency.data`. If this system property is defined then its value is the location of a properties file, the contents of which are key/value pairs of the ISO 3166 country codes and the ISO 4217 currency data respectively. The value part consists of three ISO 4217 values of a currency, i.e., an alphabetic code, a numeric code, and a minor unit. Those three ISO 4217 values are separated by commas. The lines which start with '#'s are considered comment lines. An optional UTC timestamp may be specified per currency entry if users need to specify a cutover date indicating when the new data comes into effect. The timestamp is appended to the end of the currency properties and uses a comma as a separator. If a UTC datestamp is present and valid, the JRE will only use the new currency properties if the current UTC date is later than the date specified at class loading time. The format of the timestamp must be of ISO 8601 format : `'yyyy-MM-dd'T'HH:mm:ss'`. For example,

```
#Sample currency properties
JP=JPZ,999,0
```

will supersede the currency data for Japan. If JPZ is one of the existing ISO 4217 currency code referred by other countries, the existing JPZ currency data is updated with the given numeric code and minor unit value.

```
#Sample currency properties with cutover date
JP=JPZ,999,0,2014-01-01T00:00:00
```

will supersede the currency data for Japan if `Currency` class is loaded after 1st January 2014 00:00:00 GMT.

Where syntactically malformed entries are encountered, the entry is ignored and the remainder of entries in file are processed. For instances where duplicate country code entries exist, the behavior of the Currency information for that `Currency` is undefined and the remainder of entries in file are processed.

If multiple property entries with same currency code but different numeric code and/or minor unit are encountered, those entries are ignored and the remainder of entries in file are processed.

API Note:

It is recommended to use the `BigDecimal` class while dealing with `Currency` or monetary values as it provides better handling of floating point numbers and their operations.

Since:

1.4

External Specifications

ISO - ISO 4217 - Currency codes

See Also:

`BigDecimal`, Serialized Form

Method Summary

All Methods	Static Methods	
Modifier and Type	**Method**	**Description**
`static Stream<Currency>`	**`availableCurrencies()`**	Returns a stream of available currencies.
`static Set<Currency>`	**`getAvailableCurrencies()`**	Returns a set of available currencies.
`String`	**`getCurrencyCode()`**	Gets the ISO 4217 currency code of this currency.
`int`	**`getDefaultFractionDigits`**	Gets the default number of fraction digits used with this currency.
`String`	**`getDisplayName()`**	Gets the name that is suitable for displaying this currency for the default `DISPLAY` locale.

String	**getDisplayName**(Locale locale)	Gets the name that is suitable for displaying this currency for the specified locale.
static Currency	**getInstance**(String currencyCode)	Returns the Currency instance for the given currency code.
static Currency	**getInstance**(Locale locale)	Returns the Currency instance for the country of the given locale.
int	**getNumericCode**()	Returns the ISO 4217 numeric code of this currency.
String	**getNumericCodeAsString**()	Returns the 3 digit ISO 4217 numeric code of this currency as a String.
String	**getSymbol**()	Gets the symbol of this currency for the default DISPLAY locale.
String	**getSymbol**(Locale locale)	Gets the symbol of this currency for the specified locale.
String	**toString**()	Returns the ISO 4217 currency code of this currency.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Method Details

getInstance

```
public static Currency getInstance(String currencyCode)
```

Returns the Currency instance for the given currency code.

Parameters:

`currencyCode` - the ISO 4217 code of the currency

Returns:

the `Currency` instance for the given currency code

Throws:

`NullPointerException` - if `currencyCode` is null

`IllegalArgumentException` - if `currencyCode` is not a supported ISO 4217 code.

getInstance

```
public static Currency getInstance(Locale locale)
```

Returns the `Currency` instance for the country of the given locale. The language and variant components of the locale are ignored. The result may vary over time, as countries change their currencies. For example, for the original member countries of the European Monetary Union, the method returns the old national currencies until December 31, 2001, and the Euro from January 1, 2002, local time of the respective countries.

If the specified `locale` contains "cu" and/or "rg" Unicode extensions, the instance returned from this method reflects the values specified with those extensions. If both "cu" and "rg" are specified, the currency from the "cu" extension supersedes the implicit one from the "rg" extension.

The method returns `null` for territories that don't have a currency, such as Antarctica.

Parameters:

`locale` - the locale for whose country a `Currency` instance is needed

Returns:

the `Currency` instance for the country of the given locale, or `null`

Throws:

`NullPointerException` - if `locale` is `null`

`IllegalArgumentException` - if the country of the given `locale` is not a supported ISO 3166 country code.

getAvailableCurrencies

```
public static Set<Currency> getAvailableCurrencies()
```

Returns a set of available currencies. The returned set of currencies contains all the available currencies, which may include currencies that represent obsolete ISO 4217 codes. If there is no currency available in the runtime, the

returned set is empty. The set can be modified without affecting the available currencies in the runtime.

API Note:

Consider using `availableCurrencies()` which returns a stream of the available currencies.

Returns:

a set of available currencies

Since:

1.7

See Also:

`availableCurrencies()`

availableCurrencies

```
public static Stream<Currency> availableCurrencies()
```

Returns a stream of available currencies. The returned stream of currencies contains all the available currencies, which may include currencies that represent obsolete ISO 4217 codes. If there is no currency available in the runtime, the returned stream is empty.

Implementation Note:

Unlike `getAvailableCurrencies()`, this method does not create a defensive copy of the `Currency` set.

Returns:

a stream of available currencies

Since:

25

See Also:

`getAvailableCurrencies()`

getCurrencyCode

```
public String getCurrencyCode()
```

Gets the ISO 4217 currency code of this currency.

Returns:

the ISO 4217 currency code of this currency.

getSymbol

```
public String getSymbol()
```

Gets the symbol of this currency for the default DISPLAY locale. For example, for the US Dollar, the symbol is "$" if the default locale is the US, while for other locales it may be "US$". If no symbol can be determined, the ISO 4217 currency code is returned.

If the default DISPLAY locale contains "rg" (region override) Unicode extensions, the symbol returned from this method reflects the value specified with that extension.

Implementation Requirements:

This is equivalent to calling `getSymbol(Locale.getDefault(Locale.Category.DISPLAY))`.

Returns:

the symbol of this currency for the default DISPLAY locale

getSymbol

```
public String getSymbol(Locale locale)
```

Gets the symbol of this currency for the specified locale. For example, for the US Dollar, the symbol is "$" if the specified locale is the US, while for other locales it may be "US$". If no symbol can be determined, the ISO 4217 currency code is returned.

If the specified `locale` contains "rg" (region override) Unicode extensions, the symbol returned from this method reflects the value specified with that extension.

Parameters:

`locale` - the locale for which a display name for this currency is needed

Returns:

the symbol of this currency for the specified locale

Throws:

`NullPointerException` - if `locale` is null

getDefaultFractionDigits

```
public int getDefaultFractionDigits()
```

Gets the default number of fraction digits used with this currency. Note that the number of fraction digits is the same as ISO 4217's minor unit for the currency. For example, the default number of fraction digits for the Euro is 2, while for the Japanese Yen it's 0. In the case of pseudo-currencies, such as IMF Special Drawing Rights, -1 is returned.

Returns:

the default number of fraction digits used with this currency

getNumericCode

```
public int getNumericCode()
```

Returns the ISO 4217 numeric code of this currency.

Returns:

the ISO 4217 numeric code of this currency

Since:

1.7

getNumericCodeAsString

```
public String getNumericCodeAsString()
```

Returns the 3 digit ISO 4217 numeric code of this currency as a `String`. Unlike `getNumericCode()`, which returns the numeric code as `int`, this method always returns the numeric code as a 3 digit string. e.g. a numeric value of 32 would be returned as "032", and a numeric value of 6 would be returned as "006".

Returns:

the 3 digit ISO 4217 numeric code of this currency as a `String`

Since:

9

getDisplayName

```
public String getDisplayName()
```

Gets the name that is suitable for displaying this currency for the default `DISPLAY` locale. If there is no suitable display name found for the default locale, the ISO 4217 currency code is returned.

Implementation Requirements:

This is equivalent to calling `getDisplayName(Locale.getDefault(Locale.Category.DISPLAY))`.

Returns:

the display name of this currency for the default `DISPLAY` locale

Since:

1.7

getDisplayName

```
public String getDisplayName(Locale locale)
```

Gets the name that is suitable for displaying this currency for the specified locale. If there is no suitable display name found for the specified locale, the ISO 4217 currency code is returned.

Parameters:

`locale` - the locale for which a display name for this currency is needed

Returns:

the display name of this currency for the specified locale

Throws:

`NullPointerException` - if `locale` is null

Since:

1.7

toString

```
public String toString()
```

Returns the ISO 4217 currency code of this currency.

Overrides:

`toString` in class `Object`

Returns:

the ISO 4217 currency code of this currency

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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